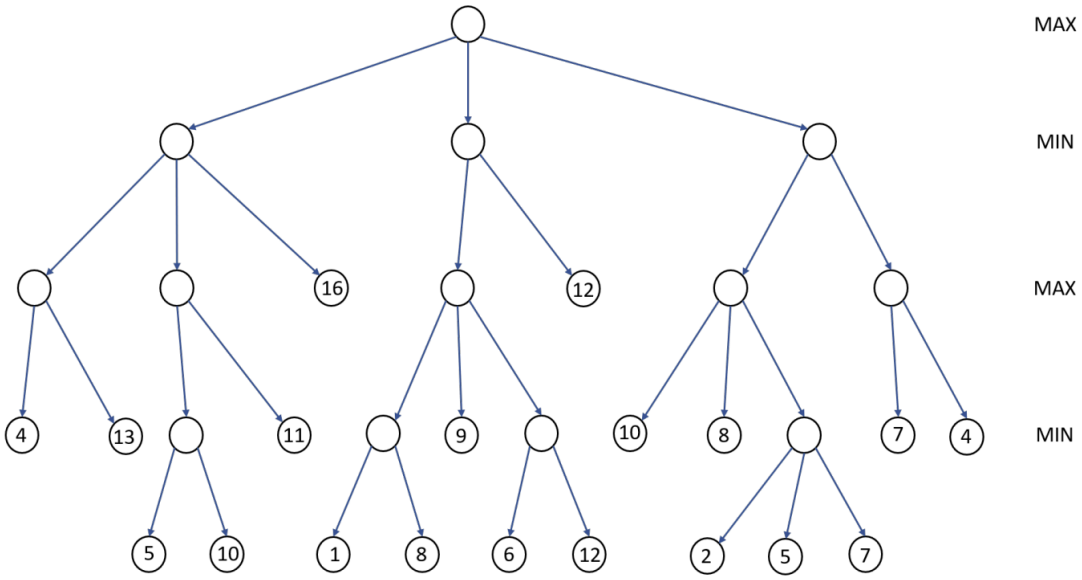
AI 500 - Practice Problem Set 1

Q1: Minimax and Alpha-Beta Pruning

a) Use the **Minimax algorithm** to compute the minimax value at each node for the game tree below.

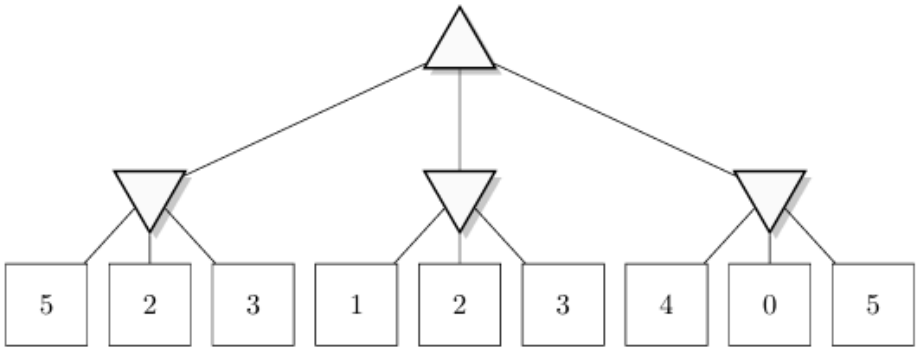


b) Use the **Alpha-Beta pruning algorithm** to prune the game tree above, assuming child nodes are visited from left to right. Show all final alpha and beta values computed at the root, each internal node visited, and at the top of pruned branches. Also show the pruned branches.

c) For a general game tree (i.e. not limited to the above tree), are there any cases in the Alpha-Beta algorithm that give a different value at the root node than the Minimax algorithm? If yes, show an example.

Q2: Minimax and Alpha-Beta Pruning

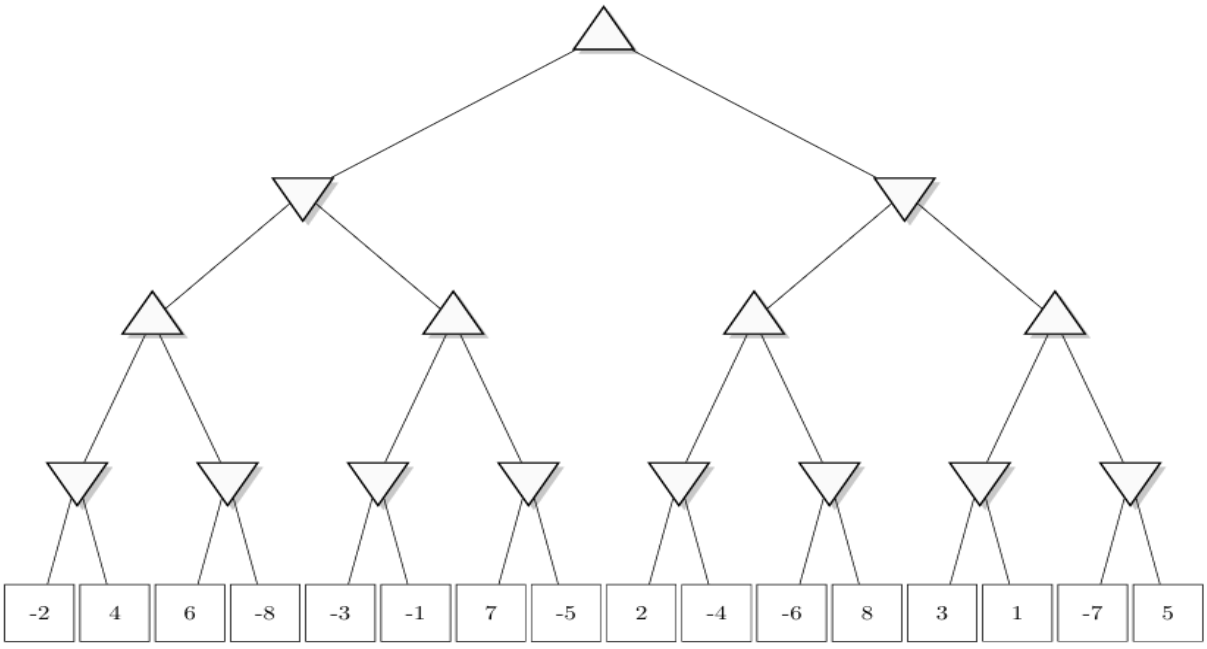
Consider the following tree representing a zero-sum game, where triangles pointing up are max nodes, triangles pointing down are min nodes, and squares are terminal nodes with the corresponding value of utility function for max player.



1. Apply the minimax algorithm for finding the best action for the max player at the root.
2. Apply the minimax algorithm with alpha-beta pruning for finding the best action for the max player at the root.

Q3: Minimax and Alpha-Beta Pruning

Consider the following tree representing a zero-sum game, where triangles pointing up are max nodes, triangles pointing down are min nodes, and squares are terminal nodes with the corresponding value of utility function for max player:



1. Apply the minimax algorithm for finding the best action for the max player at the root.

2. Apply the minimax algorithm with alpha-beta pruning for finding the best action for the max player at the root.

Question 4: Minimax

Prove the following assertion: For every game tree, the utility obtained by MAX using minimax decisions against a suboptimal MIN will never be lower than the utility obtained playing against an optimal MIN. Can you come up with a game tree in which MAX can still do better using a suboptimal strategy against a suboptimal MIN?

Question 5: RBFS

A. What is the main difference between RBFS and A* in terms of memory usage?

B. In RBFS, why do we store the "best alternative f-value" for each node?

C. Does RBFS always guarantee finding an optimal solution? Justify your answer.

Question 6: Beam Search (NLP)

i) (Beam Width = 2)

A sequence model generates tokens from the vocabulary {A, B, C, D, <END>}. At timestep 1 (after <START>), the model outputs probabilities:

- P(A) = 0.5, P(B) = 0.3, P(C) = 0.2.

At timestep 2:

- Given A: P(D|A) = 0.4, P(<END>|A) = 0.5.
- Given B: P(C|B) = 0.6, P(<END>|B) = 0.2.
- Given C: P(D|C) = 0.7, P(<END>|C) = 0.2.

Use beam search with beam width = 2.
(a) Show which beams are kept after step 1.
(b) Show which beams are kept after step 2.
(c) Which completed sequence is the top candidate?

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ii) (Beam Width = 3)

A decoder produces the following probabilities:

At timestep 1:

- P(X) = 0.4, P(Y) = 0.35, P(Z) = 0.25.

At timestep 2:

- Given X: P(Y|X) = 0.5, P(<END>|X) = 0.3.
- Given Y: P(Z|Y) = 0.4, P(<END>|Y) = 0.4.
- Given Z: P(X|Z) = 0.6, P(<END>|Z) = 0.2.

At timestep 3 (only expand non-terminated beams):

- Given XY: P(<END>|XY) = 0.7.
- Given YZ: P(<END>|YZ) = 0.6.
- Given ZX: P(<END>|ZX) = 0.5.

Beam width = 3.
(a) Show top-3 beams after timestep 1.
(b) Expand and prune after timestep 2.
(c) Which final sequence is chosen after timestep 3?

Question 7: Recursive Best First Search

Consider the Romania road map graph. The start state is ARAD and the goal state is BUCHAREST. Edge costs between cities represent driving distances in kilometers. For each city, a heuristic value h(n) is also provided. Use the RBFS algorithm to find the optimal path from ARAD to BUCHAREST.

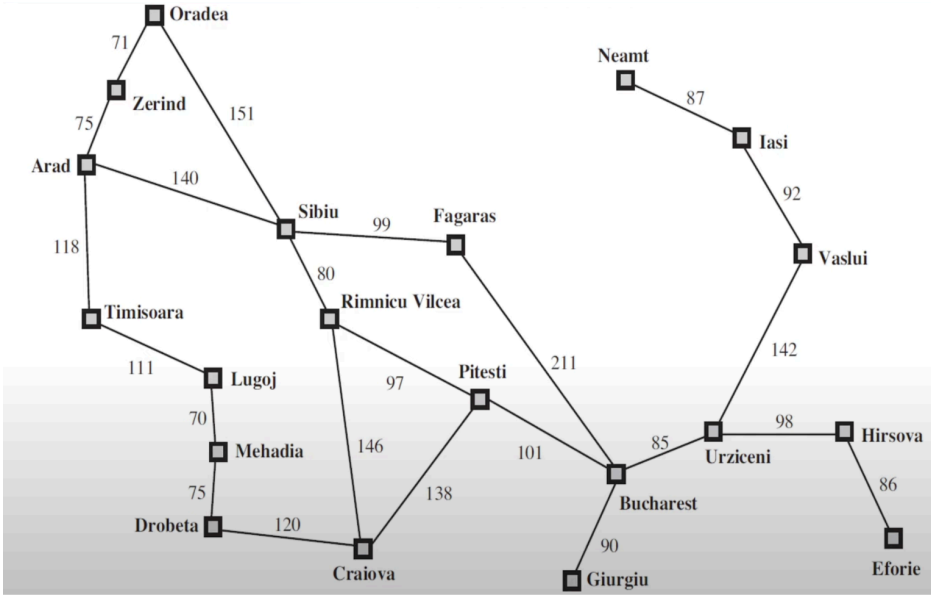
1. At each step of the algorithm, show:

- the current node being expanded,
- the f(n) values for its successors,
- the current best path being explored, and
  the alternative f-cost limit passed recursively.

2. Clearly indicate when RBFS backtracks and why.

3. Provide the final path selected by RBFS from ARAD to BUCHAREST along with the total path cost.

4. Compare the path found by RBFS to the optimal path cost (shortest distance) in the Romania map. Is the solution found by RBFS optimal in this case?



Arad	366
Bucharest	0
Craiova	160
Drobeta	242
Eforie	161
Fagaras	176
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	100
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

Question 8: Beam Search

We want to generate 3 tokens from the vocabulary:
Vocabulary: `I, You, love, hate, cats, dogs`

The model provides the following conditional probabilities:

- Start (token1):
  - *P(I)=0.55*
  - *P(You)=0.45*
- If token1 = I (token2 probabilities):
  - *P(love | I)=0.6*
  - *P(hate | I)=0.4*

- If token1 = You (token2 probabilities):
    - *P(love | You)=0.2*
    - *P(hate | You)=0.8*
- If token2 = love (token3 probabilities):
    - *P(cats | love)=0.5*
    - *P(dogs | love)=0.5*
- If token2 = hate (token3 probabilities):
    - *P(cats | hate)=0.1*
    - *P(dogs | hate)=0.9*

Using beam search with beam size = 3, generate the most probable 3-token sequence. Show the probability calculations step by step.

Question 9: Beam Search

We want to generate 3 tokens from the vocabulary:
Vocabulary: `She, He, likes, hates, pizza, sushi`

The model gives the following conditional probabilities:

- Start (token1):
    - *P(She)=0.6*
    - *P(He)=0.4*
- If token1 = She (token2 probabilities):
    - *P(likes | She)=0.7*
    - *P(hates | She)=0.3*
- If token1 = He (token2 probabilities):
    - *P(likes | He)=0.5*
    - *P(hates | He)=0.5*
- If token2 = likes (token3 probabilities):
    - *P(pizza | likes)=0.4*
    - *P(sushi | likes)=0.6*
- If token2 = hates (token3 probabilities):
    - *P(pizza | hates)=0.8*
    - *P(sushi | hates)=0.2*

Using beam search with beam size = 2, generate the top 2 sequences of length 3. Show your probability calculations step by step.

STUDY RESOURCES:

- Minimax with Alpha Beta Pruning
- Lec-18: Minimax Algorithm in Game Playing | Artificial Intelligence
- Simple Explanation of the Minimax Algorithm with Tic-Tac-Toe
- 1. Alpha Beta Pruning (Cutoff) Search Algorithm Solved Example Artificial Intelligence Ma…
- Lec-19: Alpha Beta Pruning in Hindi with Example | Artificial Intelligence

- C5W3L03 Beam Search
- How is Beam Search Really Implemented?
  - Understanding the beam search algorithm in NLP
  - Foundations of NLP Explained Visually: Beam Search, How it Works | by Ketan Doshi | TDS Archive
  - 10.8. Beam Search — Dive into Deep Learning 1.0.3 documentation

- Monte Carlo Tree Search
- Monte Carlo Tree Search - Computerphile